

Factorial Experiment Design

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Basic Factorial Design

The reason you conduct factorial experiments is to see how factors impact the response variable, as well as how each variable interacts with all combinations of the other variable. This is known as a full factorial experiment.

However, full factorial designs can be very costly and unnecessary when performing experiments on many factors. Statisticians use fractional factorial designs during early stages of experimentation for factor selection.

Sir Ronald A. Fisher wrote that complex designs, like factorial designs, are more efficient than studying each factor individually.

"Nature will best respond to a logical and carefully thought out questionnaire; indeed, if we ask her a single question, she will often refuse to answer until some other topic has been discussed."

Each factor will have 2 levels, high and low. For these notes, for some factor A, I will use A^+ for effect of A at high level.

And I will use A^- for effect of A at low level.

- Preliminary Product Process Design (PPPD) = factor screening
- After screening through potential inputs you can test more levels for each selected factor
- Need minimal experimentation
- Naive approach (randomly guessing) leads to way too many runs/trials

Steps for a Factorial Design

1. Determine the factors and levels for each factor
2. Set up way to process experimental results

k = number of factors
 l = number of levels for each factors
A 2-Factorial design will require 2^k runs

Example

Suppose you are a farmer and you want to improve wheat production. You select 2 predictors, A = water and B = sunlight, both at 2 levels, high and low. You then collect data and observe the following:

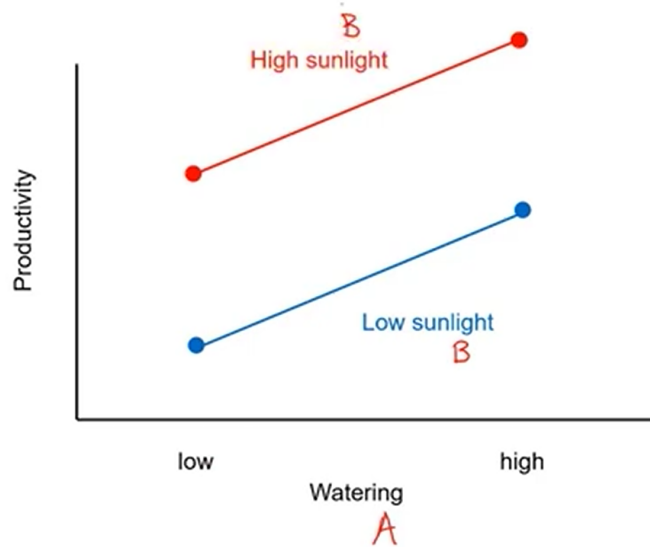


Figure 1: 2^2 Factorial Design Example